

PORTFOLIO

# Farhan Aziz Harahap.

Software Developer — Web Applications & Python Automation

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**Portfolio site** [farhan-aziz-harahap.vercel.app](https://farhan-aziz-harahap.vercel.app)

## About Me

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I'm an informatics student at Universitas Putera Batam who enjoys turning complex problems into simple, thoughtful products — from Laravel web apps to Python automation. I care about doing things properly: tested, versioned, and continuously integrated.

My journey started with a simple curiosity about how things work. Today I build database-backed web applications with Laravel and automation tools with Python. My biggest project, **Hotelaz**, is a complete hotel management system covered by 18 automated feature tests running on CI.

I'm currently deepening my testing practice and preparing for the Apple Developer Academy @ Infinite Learning (Cohort 2027) — where I plan to bring the same engineering discipline into building apps for Apple platforms.

## Skills & Tools

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### Programming

PHP · Python · JavaScript · SQL

### Web Development

Laravel · React · HTML · CSS ·  
Tailwind CSS

### Tools & Practices

Git & GitHub · GitHub Actions  
(CI) · PHPUnit · Vite · Streamlit

## Journey

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### 2023 Started learning programming

Began my informatics degree at Universitas Putera Batam and wrote my first HTML pages and simple programs.

### 2025 Built my first serious projects

Shipped my first database-backed web apps in PHP, then moved to Laravel with Hotelaz — a complete hotel management system with tests and CI.

### 2026 Automation, bots & testing

Built a Python trading-signal bot with MetaTrader 5 and Telegram, plus data tooling. Now focused on testing, CI, and writing software that holds up under real use.

## **Selected Projects** — full case studies at [farhan-aziz-harahap.vercel.app](https://farhan-aziz-harahap.vercel.app)

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## FEATURED CASE STUDY

A full-featured hotel management web app: room browsing, reservations with overbooking protection, payment-proof upload, email notifications, and an admin dashboard with revenue statistics.

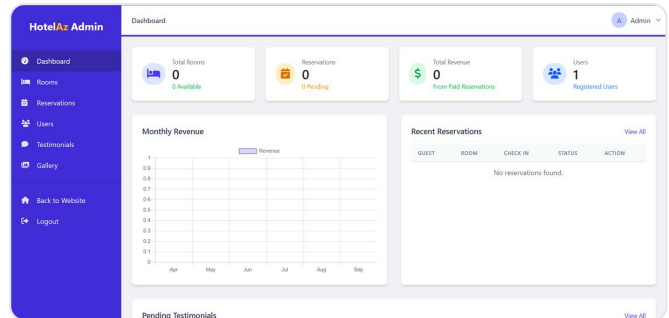
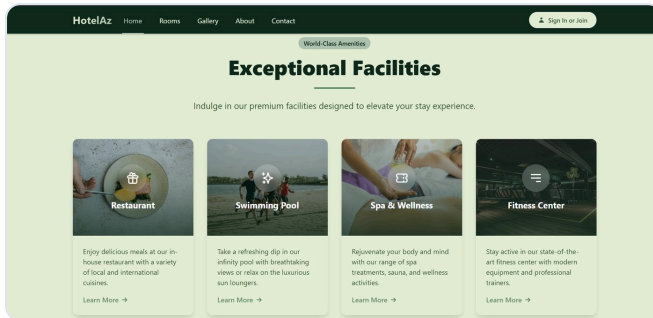
Laravel

PHP

Blade

Tailwind CSS

MySQL



## PROBLEM

Small hotel operations rely on scattered tools — paper records or spreadsheets — to manage rooms, bookings, and payments. This causes double-bookings, lost reservations, and no clear picture of revenue.

## ROLE

Solo developer — database design, backend, frontend UI, and deployment setup.

## GITHUB

sakeeeeeeeee/hotelaz

## APPROACH &amp; SOLUTION

Implemented an overlapping date-range query compared against each room's unit quantity — a booking is only accepted if active (non-cancelled) reservations for that range stay below the room count. Each room type tracks individual physical units (RoomUnit) with their own status. SMTP credentials and uploads are handled via validated storage and environment config.

## RESULT

18 automated tests (43 assertions) covering the reservation flow, availability logic, authorization, and email delivery — all passing on CI (GitHub Actions). The app sends confirmation emails with a unique booking reference (resi) generated on confirmation.

## KEY LEARNING

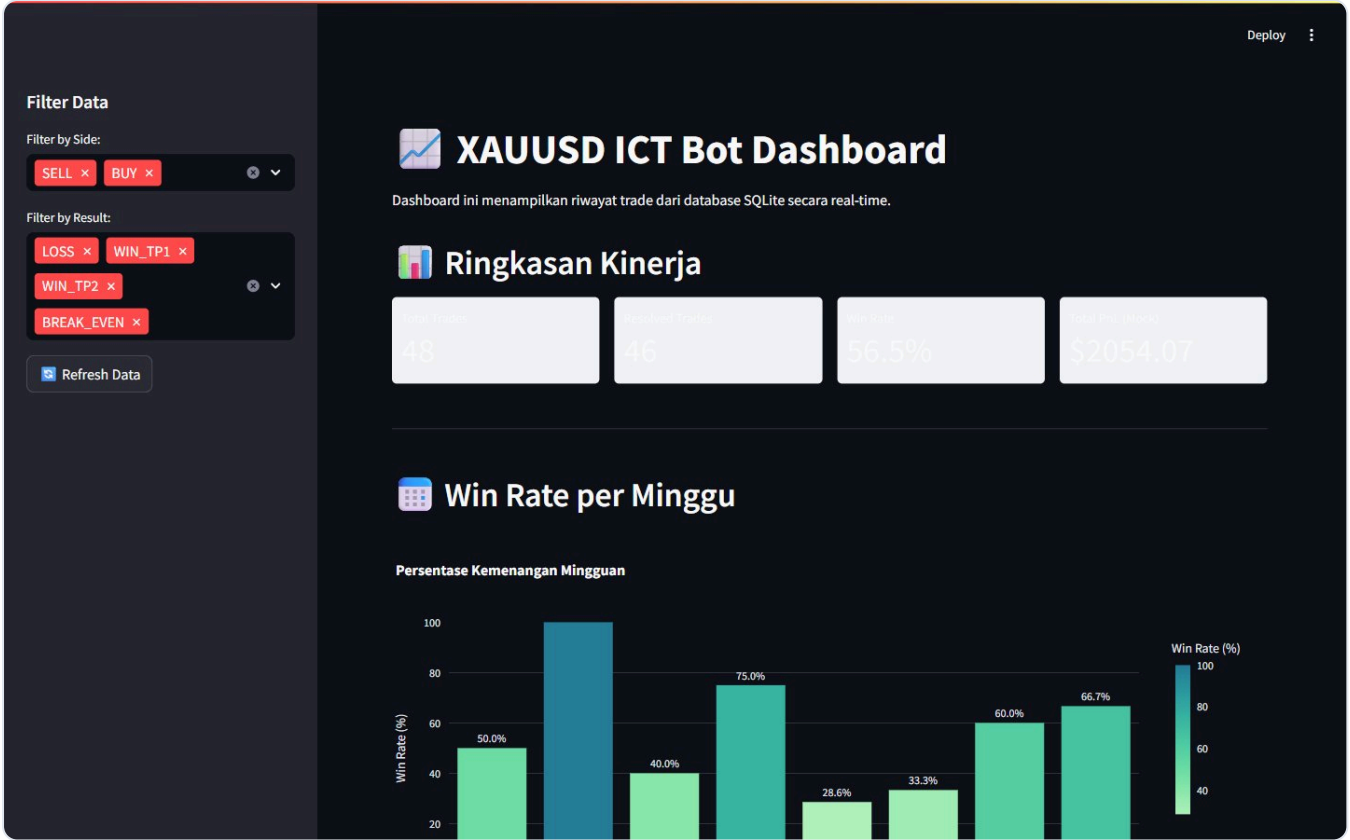
Designing correctness-critical queries (overlap detection), structuring a Laravel app with middleware-based role separation, policies for resource ownership, and testing business rules instead of just CRUD.

## NEXT STEPS

Add a payment gateway integration (the current flow records payment proofs manually), a calendar view for admins, and rate-based pricing on peak dates.

A rule-based trading signal bot for Gold (XAUUSD) implementing the ICT Silver Bullet strategy: FVG detection, session filters, smart risk management, Telegram integration, and an analytics dashboard.

- Python
- MetaTrader 5
- SQLite
- Streamlit
- Telegram Bot



PROBLEM

Manual intraday traders miss setups because monitoring charts during London and New York sessions is impractical, and emotions break consistency. Existing bots are black boxes with no risk validation.

ROLE

Solo developer — strategy engine, MT5 integration, risk manager, bot infrastructure, and dashboard.

GITHUB

sakeeeeeeee/xauusd-ict-bot

APPROACH & SOLUTION

Built a Dual-Tier architecture: the core engine runs uninterrupted on the main thread (scanning, FVG detection, SQLite persistence) while the Telegram interface answers commands asynchronously in a background thread. Risk manager computes ATR-based stop loss, enforces R:R bounds, and validates min/max risk. A backtest engine with walk-forward train/test splits and an offline Gemini-analysis script (weekly, manual) complete the loop — LLMs are explicitly excluded from live execution.

RESULT

A complete pipeline: MT5 data → signal detection → risk validation → Telegram broadcast with charts → outcome tracking in SQLite → Streamlit analytics showing historical win rate. Covered by unit tests for risk and analysis modules.

#### KEY LEARNING

Translating a discretionary trading methodology into deterministic code, walk-forward testing to fight overfitting, threading and async design for responsive bots, and separating measurable evaluation from execution.

#### NEXT STEPS

Expand to multi-symbol scanning, add proper position-sizing from account equity, and publish the backtest reports as static pages.

## **Selected Projects** (continued)

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## DATA &amp; AUTOMATION

A Python utility that collects posts matching a keyword with cookie-based account rotation, exporting results to CSV and JSON for analysis.

Python

Twikit

CSV

JSON

```
Microsoft Windows [Version 10.0.22631]
C:\...\scrape_x> dir /b
.gitignore
config.example.json
README.md
requirements.txt
scraping.py

C:\...\scrape_x> type config.example.json
{
  "cookies_list": [
    {
      "auth_token": "ISI_AUTH_TOKEN_AKUN_1",
      "ct0": "ISI_CT0_AKUN_1"
    },
    {
      "auth_token": "ISI_AUTH_TOKEN_AKUN_2",
      "ct0": "ISI_CT0_AKUN_2"
    }
  ],
  "keywords": ["Prabowo"],
  "target_tweets": 500,
  "tweets_per_account": 250,
  "delay_between_requests": 10,
  "max_tweets_per_request": 100,
  "max_retry_on_duplicate": 5
}

C:\...\scrape_x> python scraping.py
2026-09-12 20:13:54,816 - config.json tidak ditemukan. Salin config.example.json ke config.json lalu isi cookies.

C:\...\scrape_x> _
```

## PROBLEM

Researchers and analysts need bulk access to public posts matching specific keywords, but exporting them manually is slow and the official API is costly.

## ROLE

Solo developer — design, implementation, and testing.

## GITHUB

sakeeeeeeeee/scrape\_x

## APPROACH &amp; SOLUTION

Implemented automatic cookie-based account rotation — when one session hits a limit, the scraper transparently switches to the next account and continues. Results are deduplicated and exported to both CSV and JSON.

## RESULT

A working command-line tool that collects keyword-matched posts continuously without manual intervention, used for personal research datasets.

## KEY LEARNING

## NEXT STEPS

Practical web-data ethics and rate-limit handling, session management, and designing for failure (resumable runs, structured logging).

Add keyword scheduling, media download support, and a simple summary report of collection statistics.

## LEARNING IN PUBLIC

A PHP CRUD application for managing school data — the project where I first connected forms, databases, and server-side validation into one working system.

PHP

MySQL

CRUD

## DATA SISWA

TAMBAH DATA

| NO. | NISN       | NAMA LENGKAP        | ALAMAT                            | AKSI |       |
|-----|------------|---------------------|-----------------------------------|------|-------|
| 1   | 0045871234 | Ahmad Rizki Pratama | Jl. Sudirman No. 12, Batam Center | EDIT | HAPUS |
| 2   | 0045871235 | Siti Nurhaliza      | Jl. Ahmad Yani No. 45, Batam      | EDIT | HAPUS |
| 3   | 0045871236 | Budi Santoso        | Jl. Raya Kabil No. 8, Nongsa      | EDIT | HAPUS |
| 4   | 0045871237 | Dewi Lestari        | Jl. Imam Bonjol No. 30, Sekupang  | EDIT | HAPUS |
| 5   | 0045871238 | Muhammad Farhan     | Jl. Diponegoro No. 7, Batu Aji    | EDIT | HAPUS |

## PROBLEM

School administrative data (students, classes, records) is often managed on paper, with no validation and no history.

## ROLE

Solo developer — full stack, from schema to UI.

## GITHUB

sakeeeeeeeee/CRUD-form-sekolah

## APPROACH &amp; SOLUTION

A straightforward PHP + MySQL application with prepared statements, server-side validation on every write operation, and clean feedback for each action.

## RESULT

A functional data-management app that became the foundation for my later Laravel work — the same concepts, now with a framework.

## KEY LEARNING

## NEXT STEPS

Core CRUD mechanics, SQL injection awareness (prepared statements), and why server-side validation is non-negotiable.

Rebuild it in Laravel as a comparison exercise to show the same problem solved at a higher abstraction level.

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## Highlights & Activities

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### **Apple Developer Academy @ Infinite Learning — Applicant, Cohort 2027**

Batam, Indonesia · 2026

Preparing through Apple ecosystem learning paths while finishing my informatics degree.

### **Open-source portfolio on GitHub**

[github.com/sakeeeeeeeee](https://github.com/sakeeeeeeeee) · 2026

Web apps, automation tools, and a tested trading bot with continuous integration.

### **Test-driven & CI workflows**

Self-directed · 2026

Added automated test suites and GitHub Actions pipelines to my projects — 18 feature tests on my main web app.